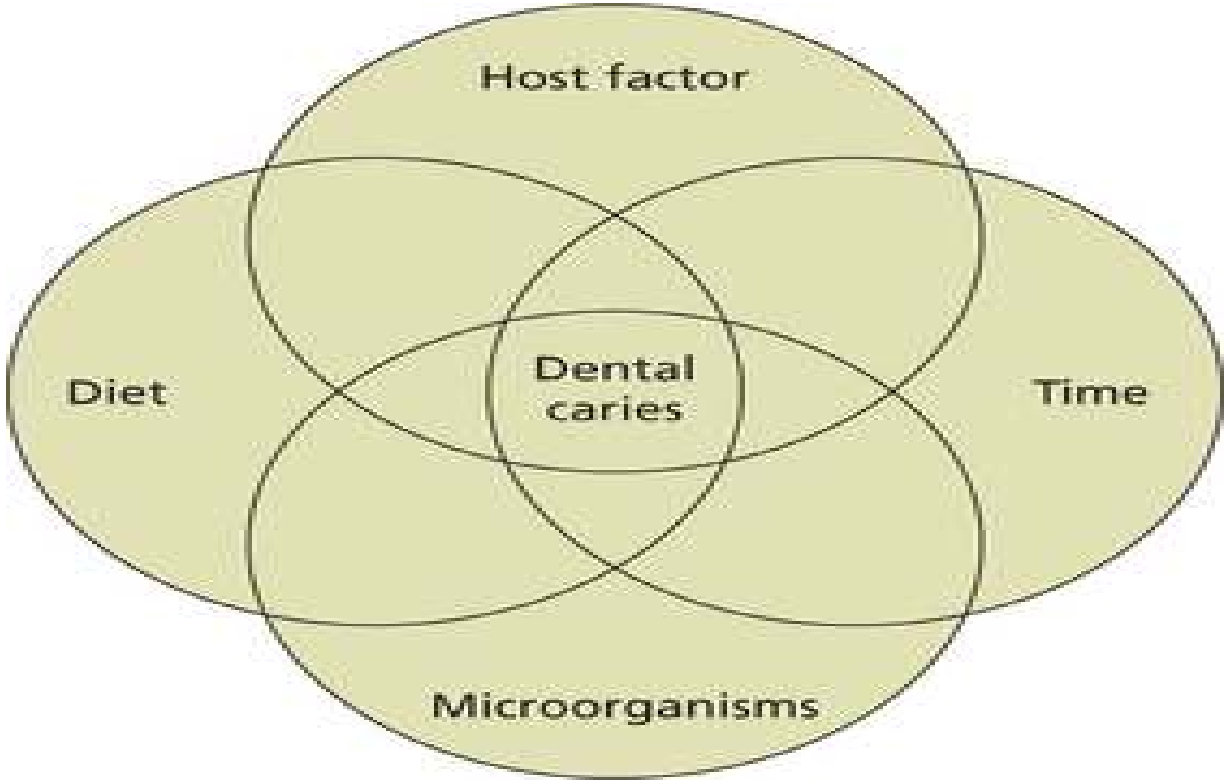
ORAL PATHOLOGY
Dental Caries

- **In general; there are two ways to examine dental caries under microscope.**
- **Ground section** is the first way. No pulp present in ground section, thus it is mainly for studying enamel.
- **Decalcified section** can be prepared as the second way, here enamel is lost during decalcification, and thus it is mainly for studying pulp and other structures.

Introduction

- Dental caries is a multifactorial disease.
- Dental caries is the most prevalent chonic disease in man , 95% of population have a decay or will have it before the die.



- **Dental caries can be controlled by controlling the <u>four</u> main factors that are related to it: -**

1- Host (tooth): Morphology,Position,Fluoride , Age

2- Saliva: Antibacterial, Flow rate, PH ,Viscosity

3- Diet (food): sucrose is considered as the most cariogenic type of sugar because (1) readily fermented by bacterial plaque (2) its easily converted to extracellular Glucans which act as glue for bacteria helping their adherence to tooth surface.

4- Plaque and cariogenic flora: Plaque considered as a contributing factor for at least initiation of caries. However mere presence of dental plaque doesn't mean caries will occur.

Etiology and pathogenesis:

Etiology is still controversial and not clear.

- **Acidogeneic theory (miller s chemoparasitic theory 1890):-** it is the most accepted and supported theory showed that in germ free oral hygiene in some laboratory animals ,even with administration of sugar; there is **no dental caries** in these animals. Thus, dental caries is produced by chemical action of acids produced by micro flora.
- Miller s theory suggests that dental caries develop in two phases.
- In the **f ırst phase,** micro f bra attack the inorganic structure, where decalcif cation of enamel and dentin is carried out by acids produced as a result of fermented sugar accumulating on

2) proteolytic theory (bodecker 1878) :-

Main suggestion of this theory is that microorganism attack the organic part of enamel, leaving the generated acid responsible for further decalcification of inorganic part .

Bodecker suggested that bacteria could penetrate into enamel through lamellae and interprismatic substance.

- ***Objections to the theory:*** - out of 0.56% of organic matrix, 0.18% is keratin. However, no enzyme systems capable of attacking keratins have been isolated so far.
- Studies in germ free rats have shown that caries can occur in the absence of proteolytic organisms.

- ***3-proteolysis – chelation theory:-***
- Schatz et al in 1955 proposed that the product of proteolysis of the tooth substance, acquired pellicle and foot substance act as **chelating agents** which remove calcium ions from tooth by forming complex. This process could account for caries activity at neutral pH. Objection to the theory that the recent studies have shown that saliva as well as plaque does not contain substance in suf ficient concentrations to chelate calcium from enamel .

- ***Clinical classification of dental caries:-***

whether lesion is new or under previous restoration: -

1. Primary caries 2. Secondary caries

According to site of attack, it is classified as follows:

- 1. Pit and fissure caries. 2. Smooth surface caries. 3. Cemental or root caries.
- **According to rate of attack, dental caries is classified in to:**
- 1. Rampant or acute caries. 2. Slow progressive or chronic caries.
- 3. Arrested caries.

1.pit and fissure caries

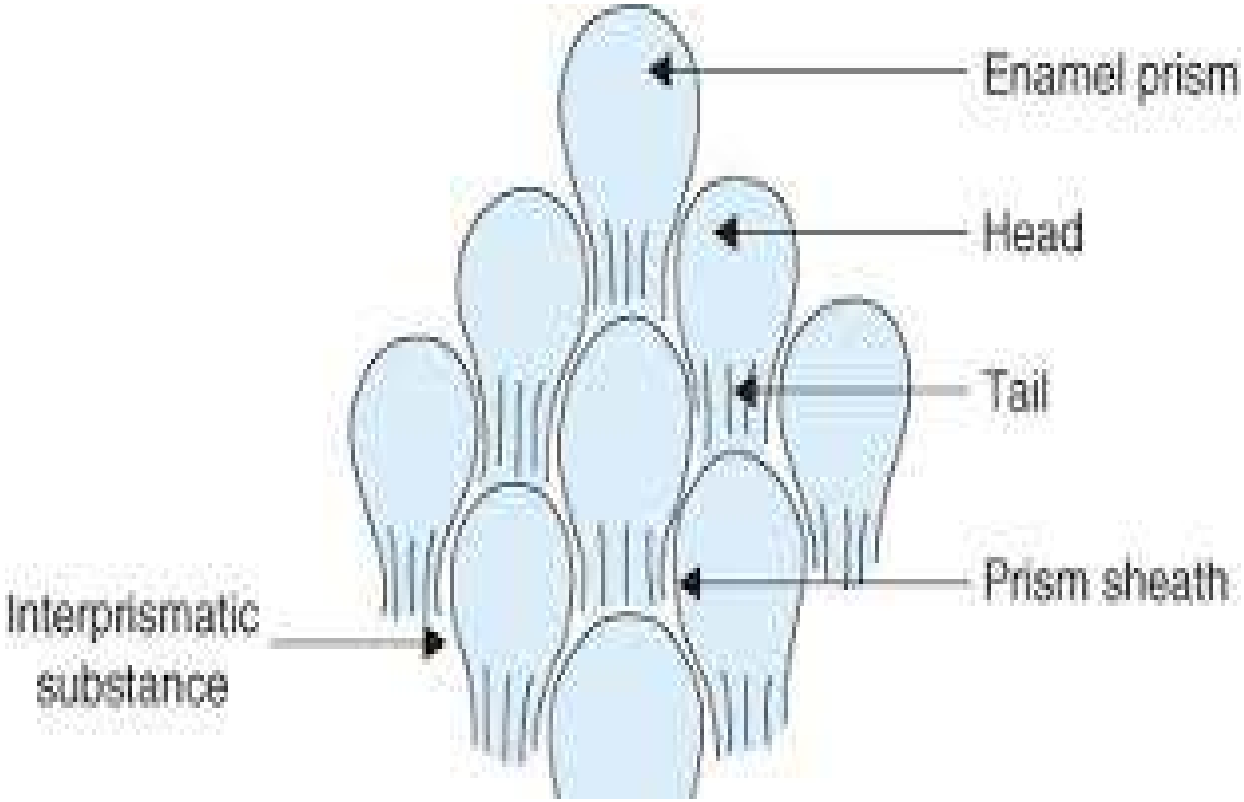
Clinically:

- Black or brown discoloration in the pits and fissure.
- When inspected with probe the probe stick to it.
- When caries spread laterally in dentin enamel appear chalky white (undermined).

Dental histology

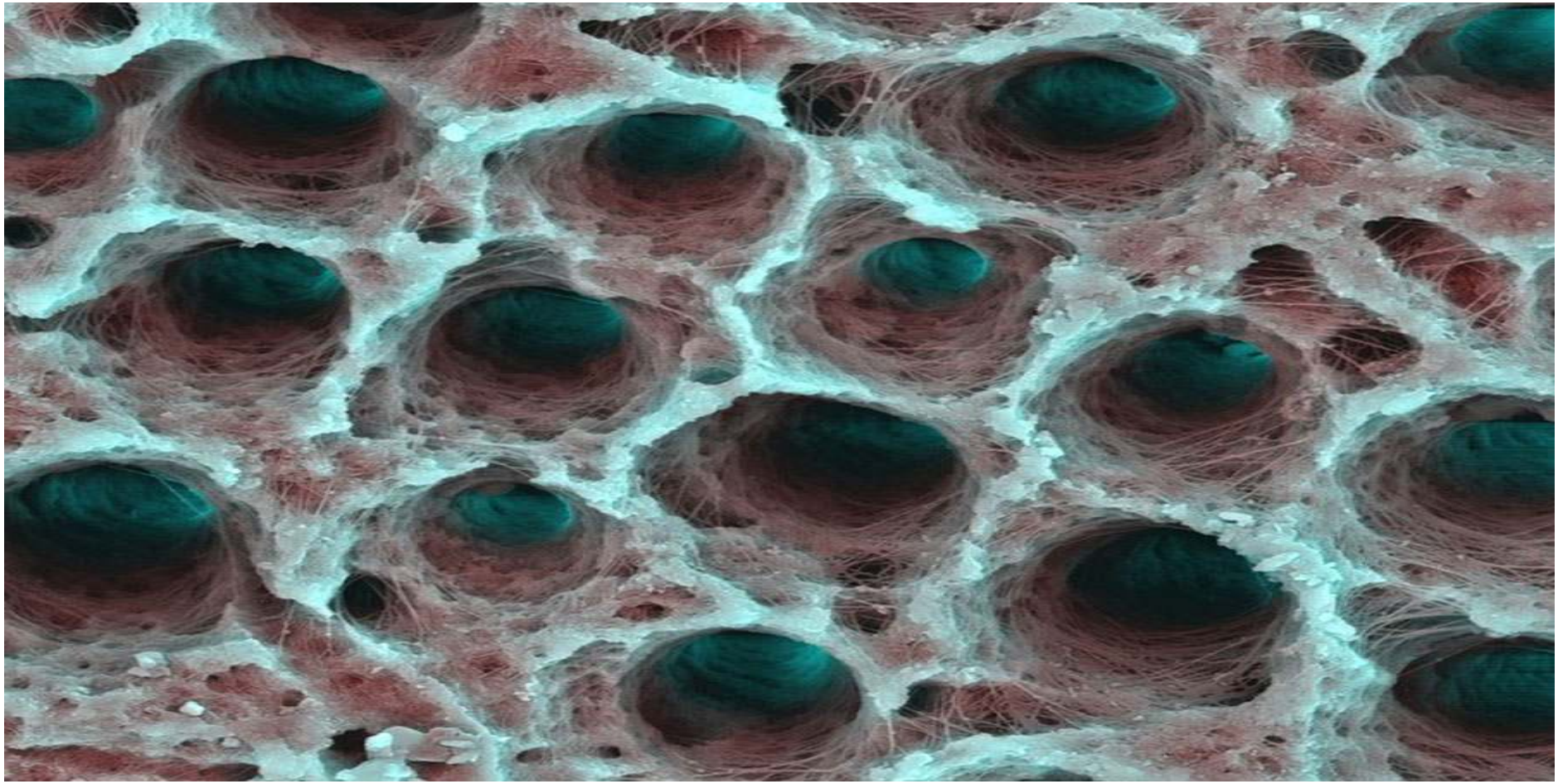
Enamel

- Composed of 96% inorganic material 4% organic and water.

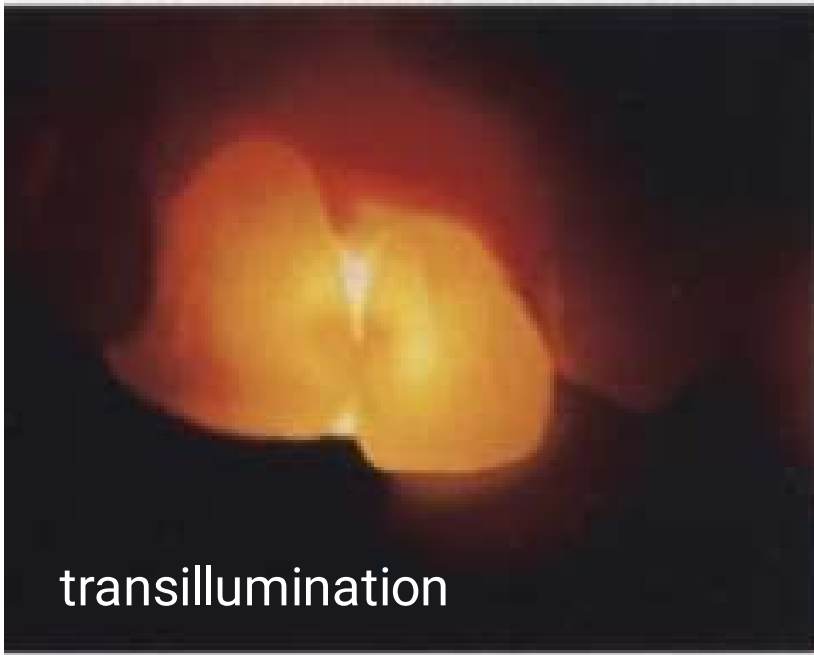


Dentin

- Composed 30% organic material and 70% inorganic material.
- Types??



Class III interproximal decay of incisors that appears as a brownish triangle just below the contact point



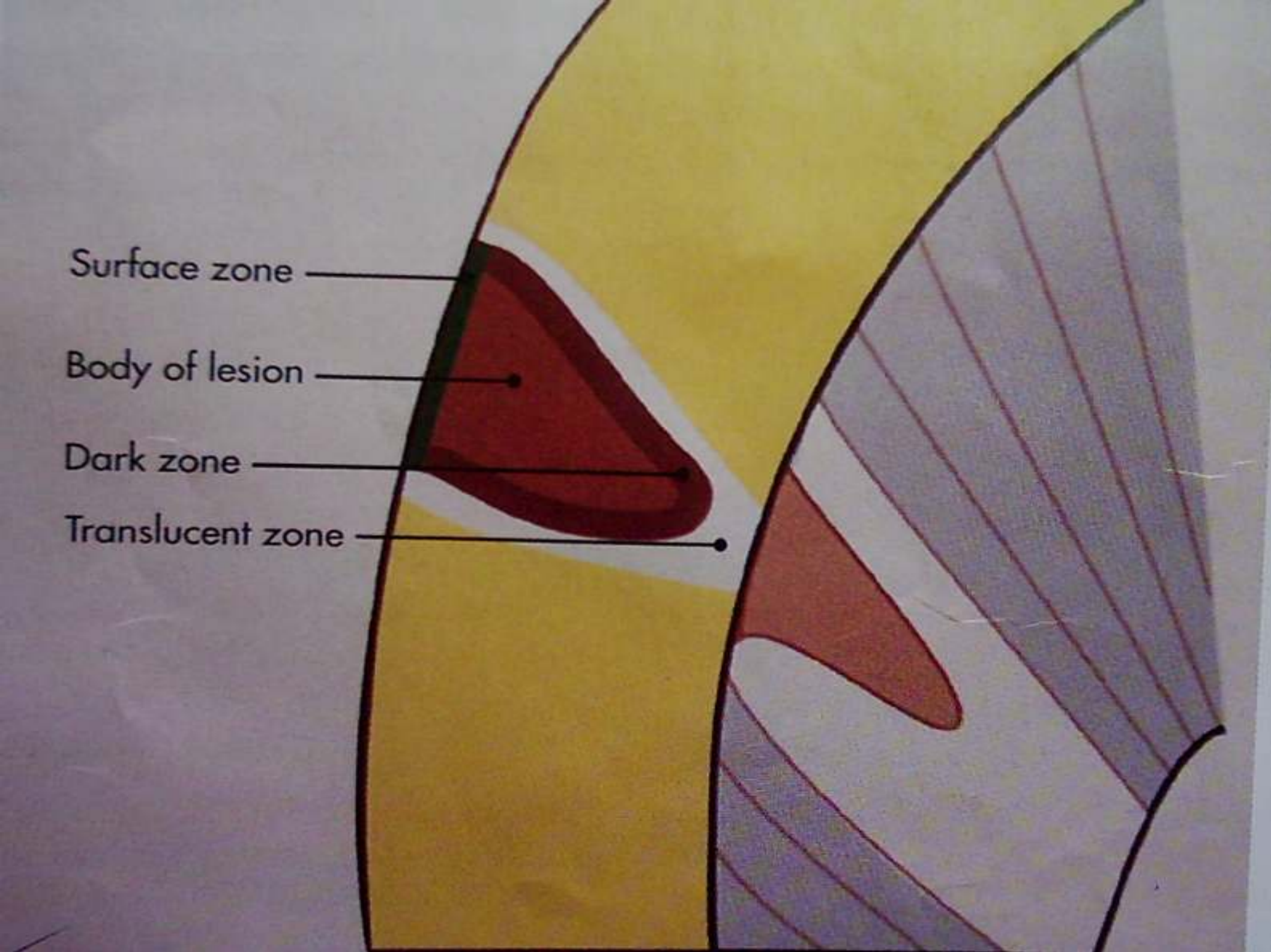
gure 17.3. Rampant caries associated with xerostomia.

1. A tooth surface without caries.
2. The first signs of demineralization.
3. The enamel surface has broken down.
4. A filling has been made but the demineralization has not been stopped.
5. The demineralization proceeds and undermines the tooth.
6. The tooth has fractured.

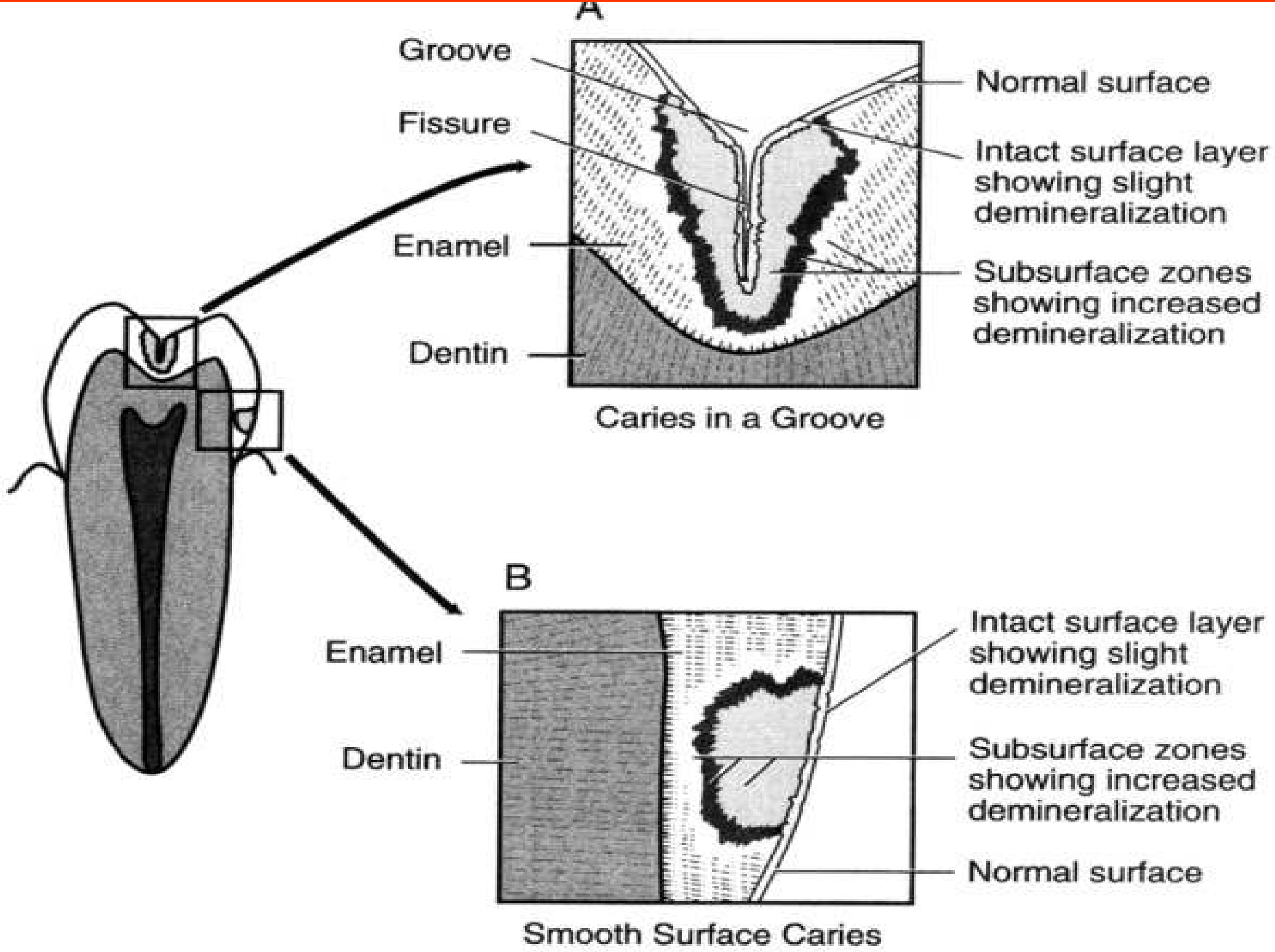
Several zones can be distinguished before complete destruction of the enamel

- **i: translucent zone** :In this zone, demineralization has taken place ,lies at the advancing front of the lesion. More porous than sound enamel. Pores have been created by the demineralization process.
- **ii: dark zone** Some remineralization happens due to reprecipetation of minerals, lost from the translucent zone just superficial to the translucent zone. More porous than the translucent zone.
- **iii: body of the lesion** :It is the area of greatest demineralization ,The largest portion of the lesion superficial to the dark zone. Increase in porosity from the peripheries to the center.
- **iiii: the surface zone:** unaffected surface layer that cover the small lesion. High degree of mineralization than subsurface enamel.

if the lesion progress the surface layer will be destroyed a cavity forms.


Surface zone
Body of lesion
Dark zone
Translucent zone

- **Note:- more cavitation's in pits and fissure than smooth surface caries. Why?**
- The enamel at the bottom of the pit and fissure may be very **thin**, so caries reach faster.
- In pits and fissure, the enamel rods are directly directed **laterally** (diverge), when caries occur it follows the direction of these rods leading to the formation of cone shaped or triangle.
- The enamel at the surface become **undermined** and starts to collapse under the stress of mastication and to fragment around the edge of the cavity. By this stage, bacterial attack on the dentin is well established.


A
Groove
Fissure
Enamel
Dentin
Normal surface
Intact surface layer showing slight demineralization
Subsurface zones showing increased demineralization
Caries in a Groove
B
Enamel
Dentin
Intact surface layer showing slight demineralization
Subsurface zones showing increased demineralization
Normal surface
Smooth Surface Caries

- **At early stage of dentin caries, a defensive mechanism of dentinal tubule and the vital pulp occur by:**

1-Development of dentinal sclerosis or translucent dentin which mean calcification of dentinal tubules which will seal them to prevent bacterial penetration, this form in a band about half way between the pulp and adj.

2- the odontoblast in the pulp react to changes in dentin by formation of reparative dentin (**tertiary dentin – a tubular dentin**) this dentin is localized to the irritant odontoblast irregular or a tubular dentin.

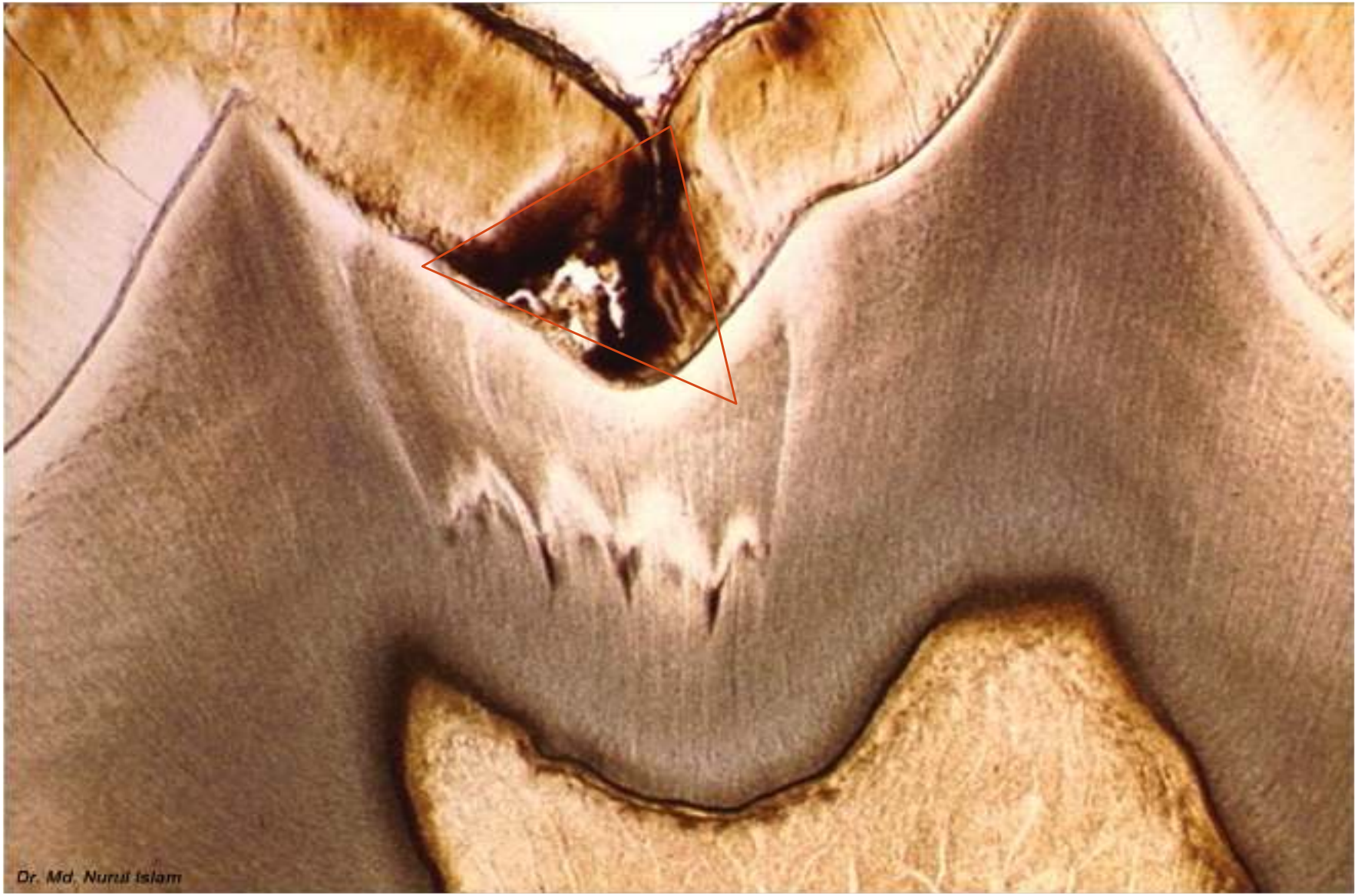
3- secondary dentin: tubular dentin separated from primary dentin by hyperchromatic line or demarcated zone. It is formed following

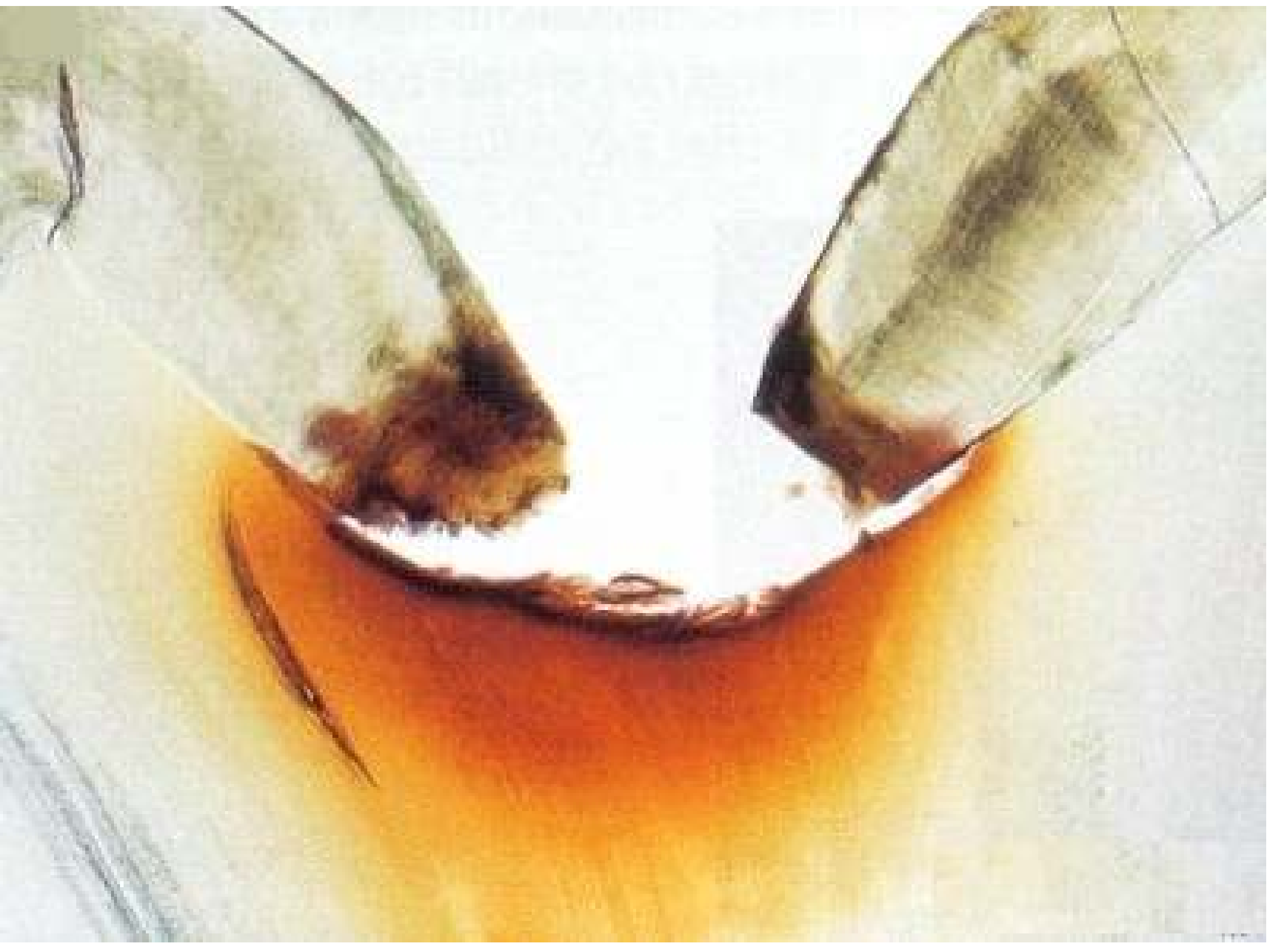
- **Arrested caries** and remineralization precavity (Enamel white spot) may become arrested when the adjacent tooth is removed so that the stagnation area is removed, the lesion may become remineralized by mineral from the saliva.
- **Dentin caries** may occasionally be **arrested** as a result of destruction of so much enamel, that a wide area of dentin become exposed, if this surface is then subject to attrition , plaque deposition may be prevented by use of f luoride and consumption of a less cariogenic diet may cause a surface lesion in enamel to heal

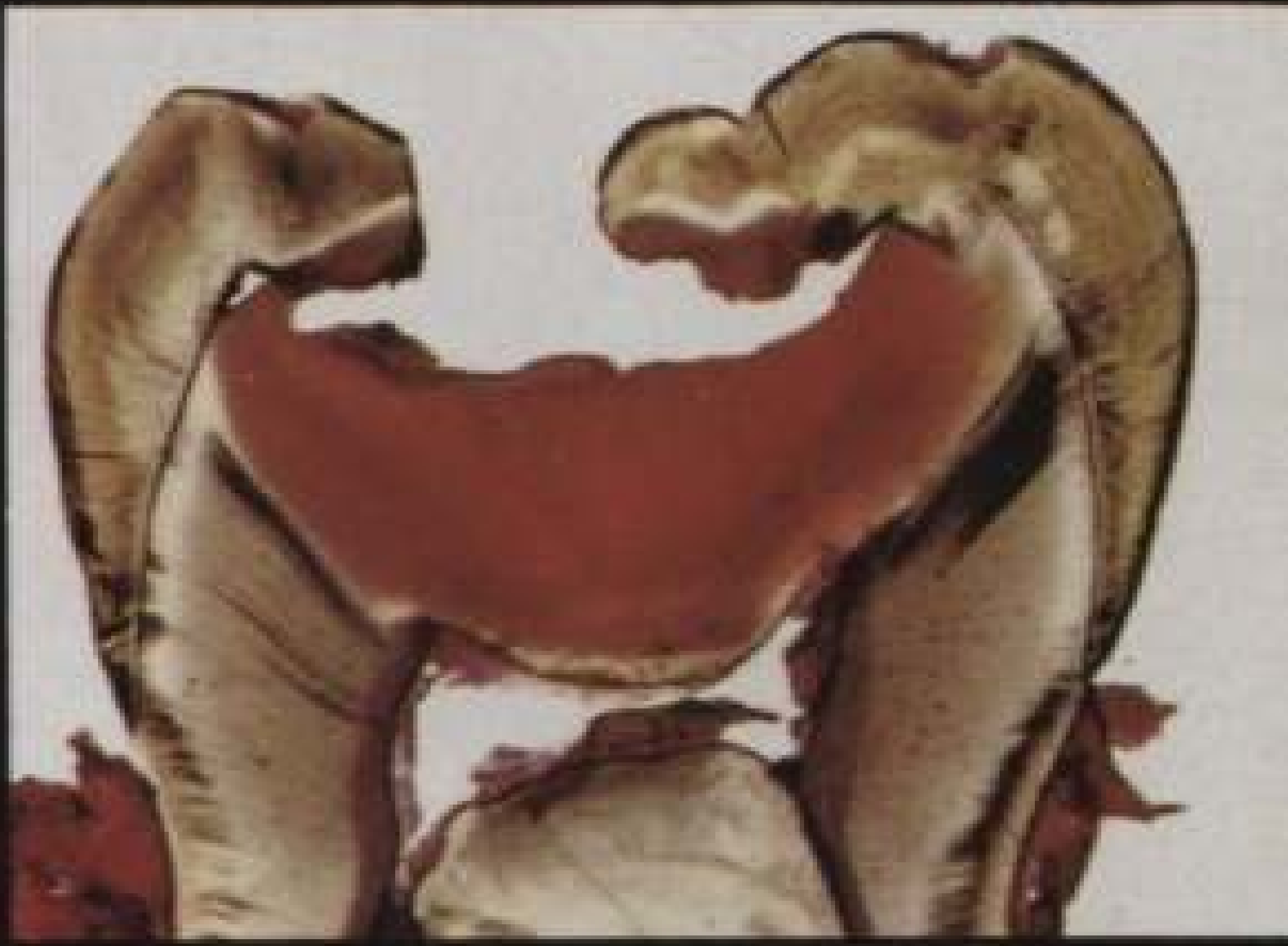
Caries process involve???

Histopathlogically

enamel







2.Smooth surface caries.

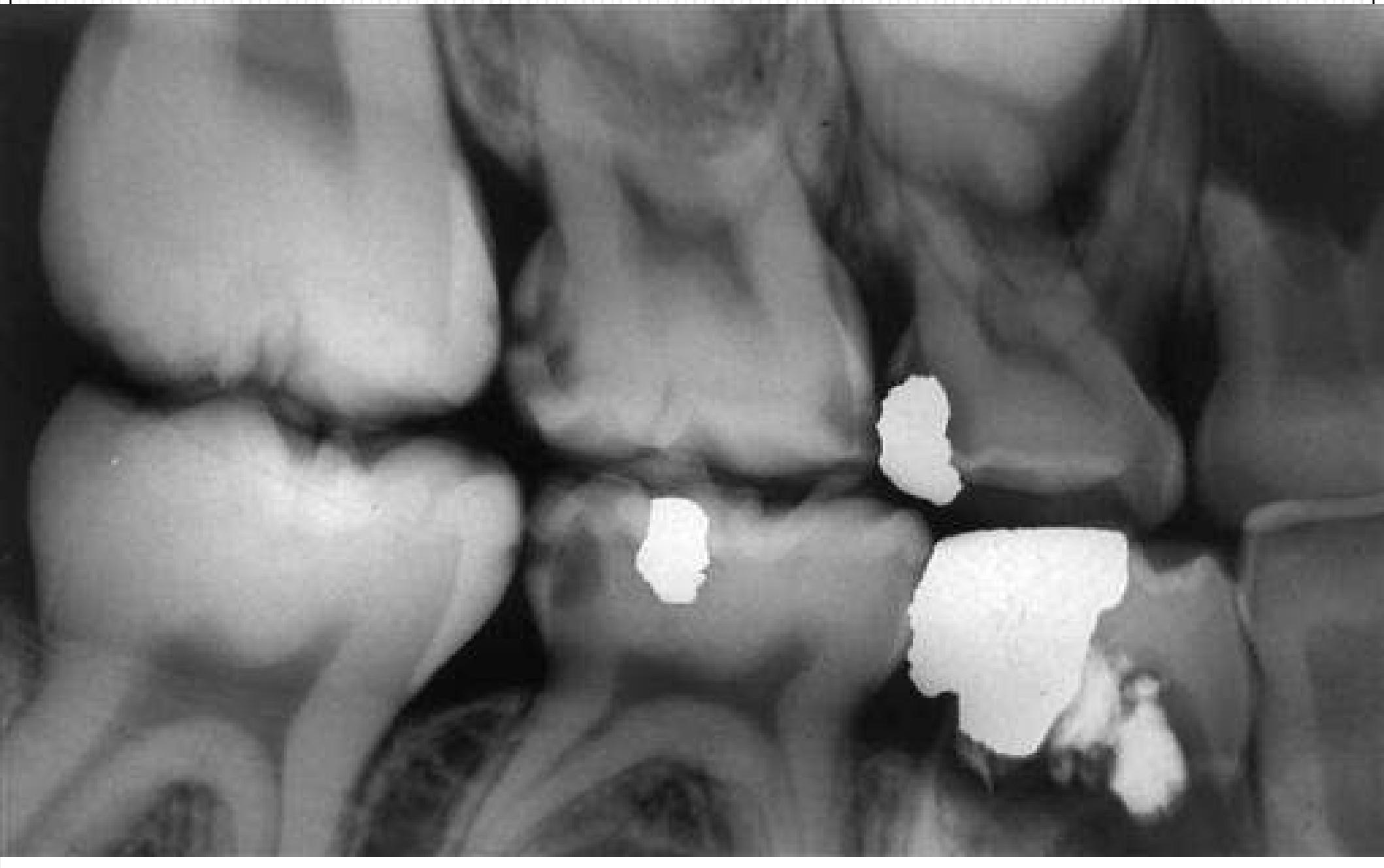
Clincally:

- Frequently on proximal surface and gingival third of buccal and lingual surfaces
- Proximal caries occur just below the contact point and appear as chalky white well-demarcated opacity of enamel.

- Histopathologically

enamel

Radiographically:

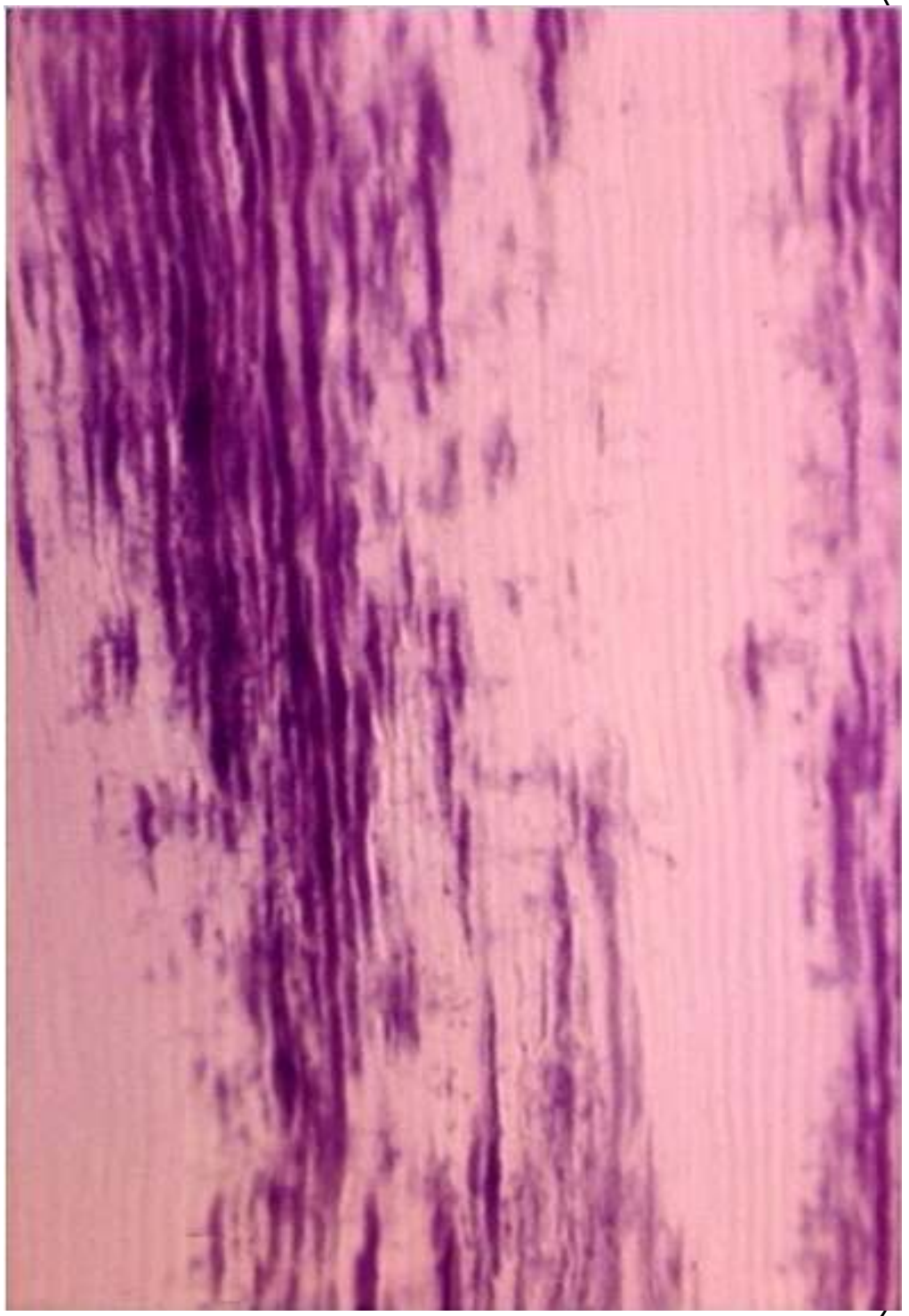


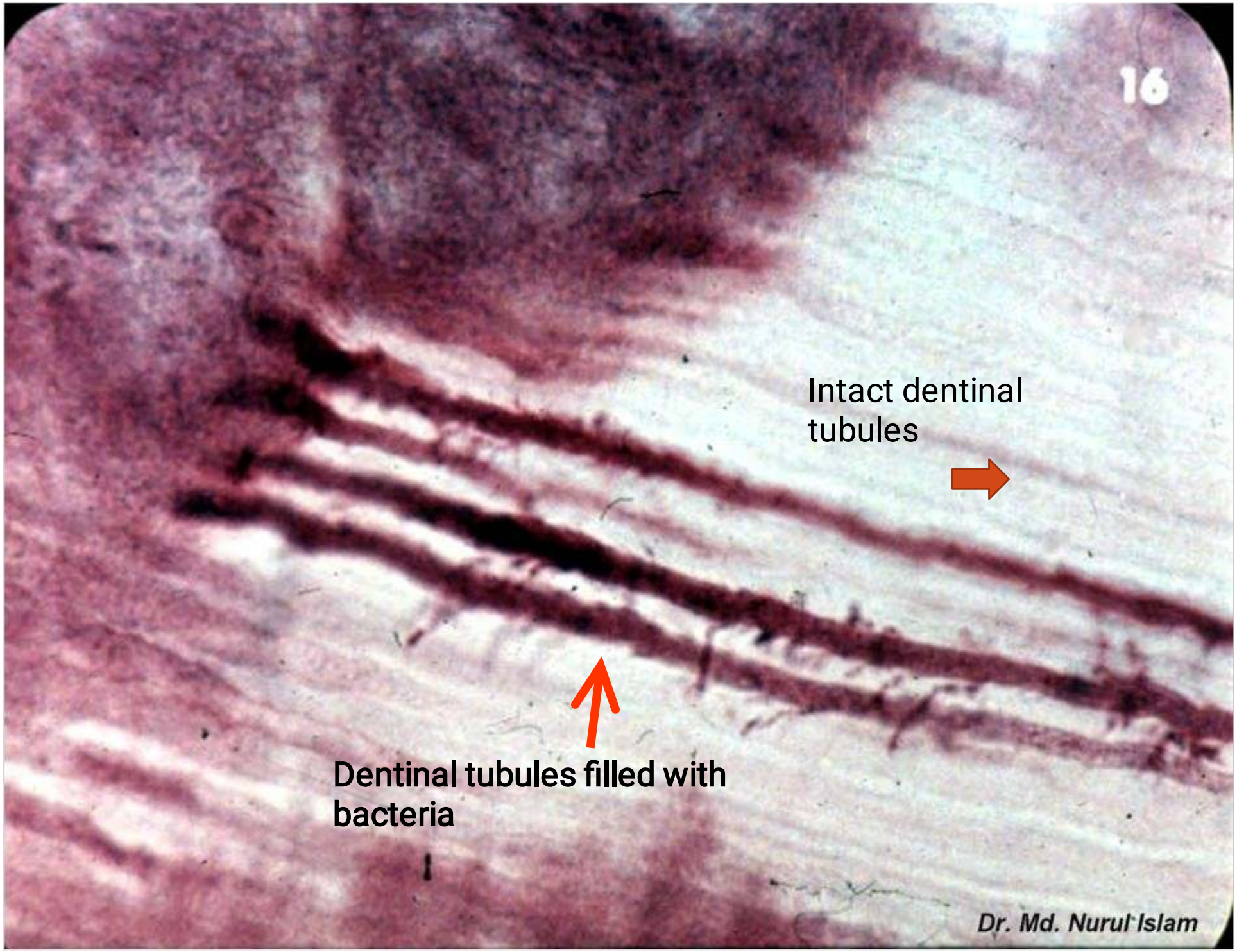
Caries of dentin

- Begins with the natural spread of the process along the DEJ and rapid involvement of the dentinal tubules.
- The dentinal tubules act as tracts leading to the pulp (path for micro-organisms).

- **Early Dentinal Changes:**
- -initial penetration of the dentin by caries→ dentinal sclerosis, calcification of dentinal tubules and sealing off from further penetration by micro-organisms, more prominent in slow chronic caries.

This initial decalcification involves the walls allowing them to distend as the tubules are packed with microorganisms. Each tubule is seen to be packed with pure forms of bacteria, eg., one tubule packed with cocci forms the other tubule with bacilli.

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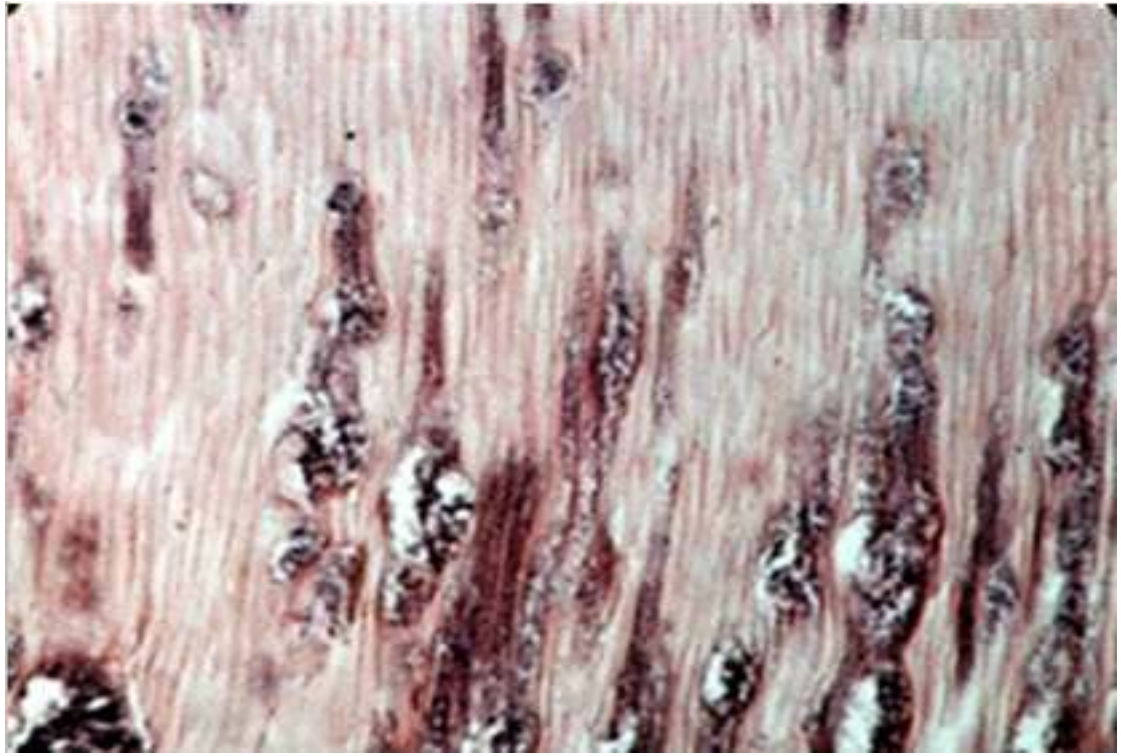
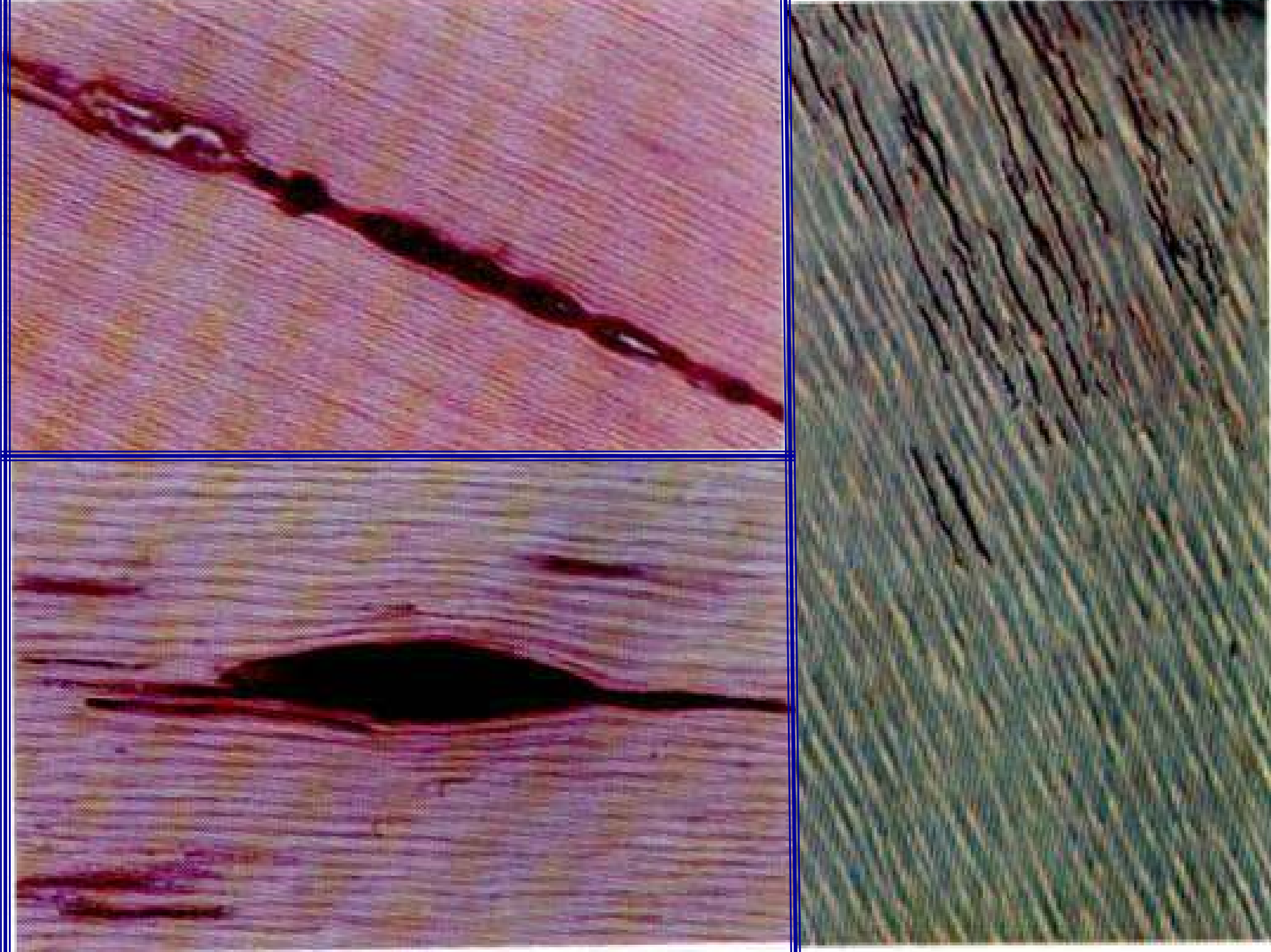
Intact dentinal tubules
Dentinal tubules filled with bacteria
Dr. Md. Nurul Islam

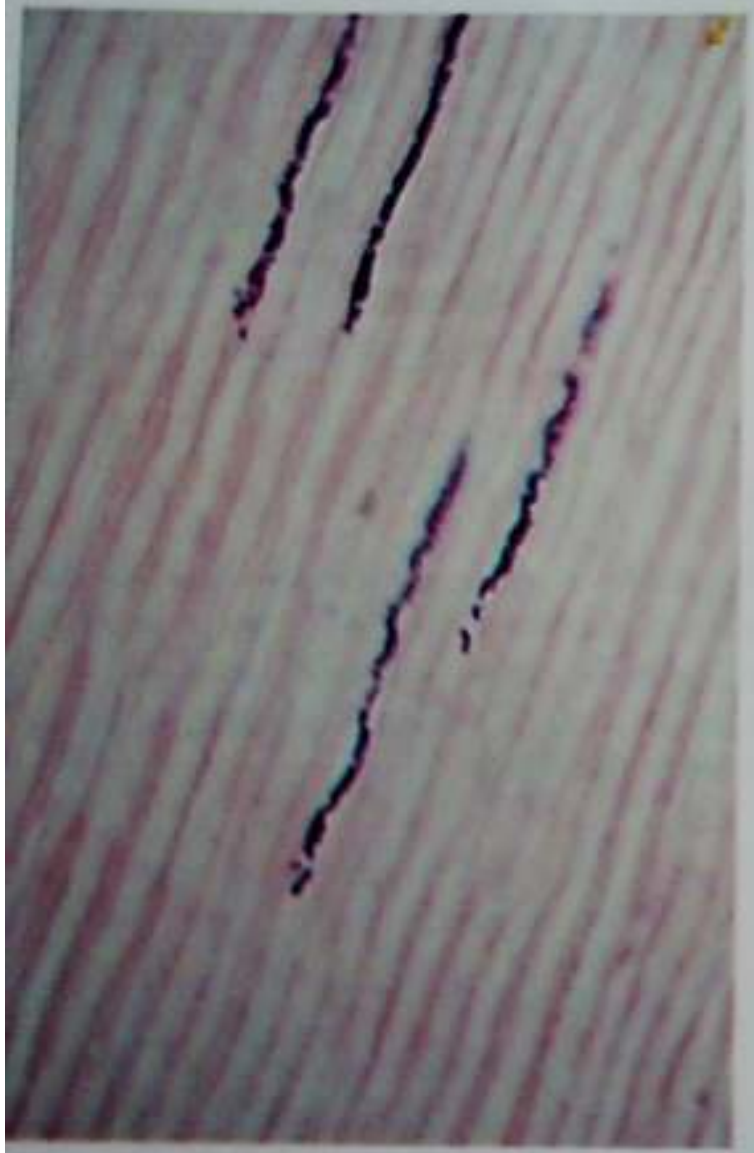
Advanced Dentinal Changes ;

•decalcification of walls, confluence of the dentinal tubules, -tiny "liquefaction foci", described by Miller are formed by the focal coalescing and breakdown of dentinal tubules.

•These are ovoid areas of destruction parallel to the course of the tubules which filled with necrotic debris and increase in size by expanding. The adjacent tubules are distorted and their course is bent due to this expansion.



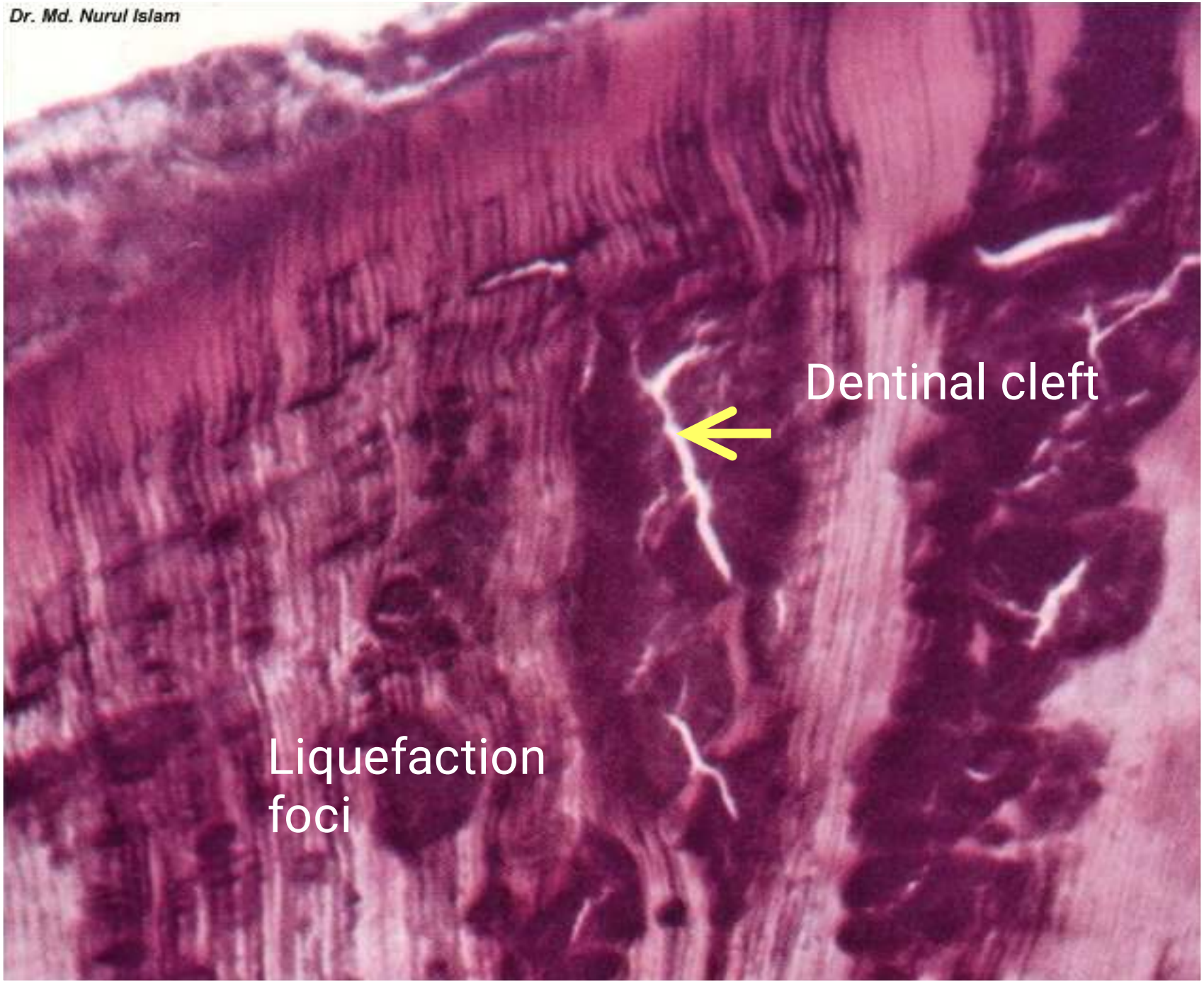


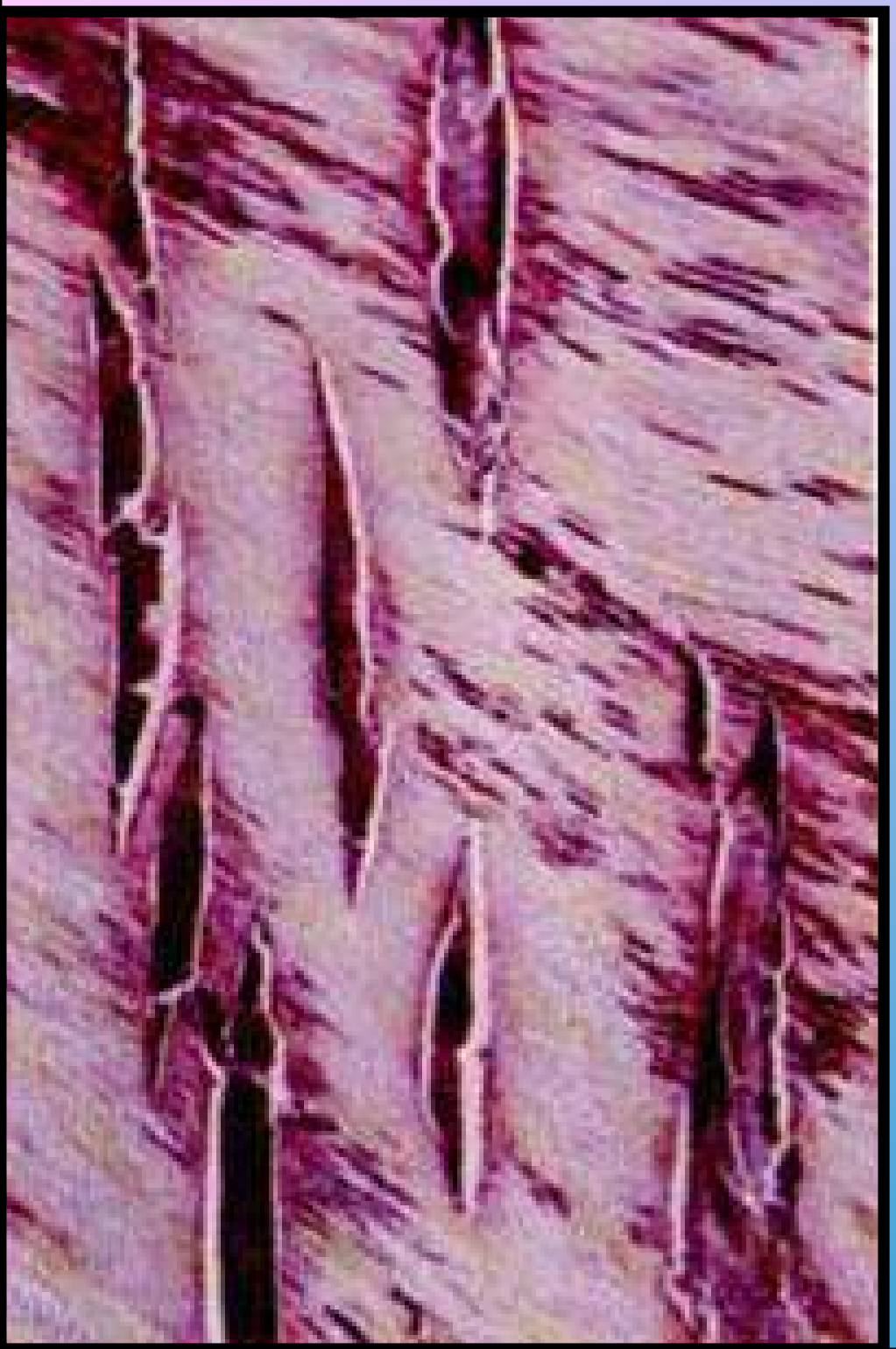


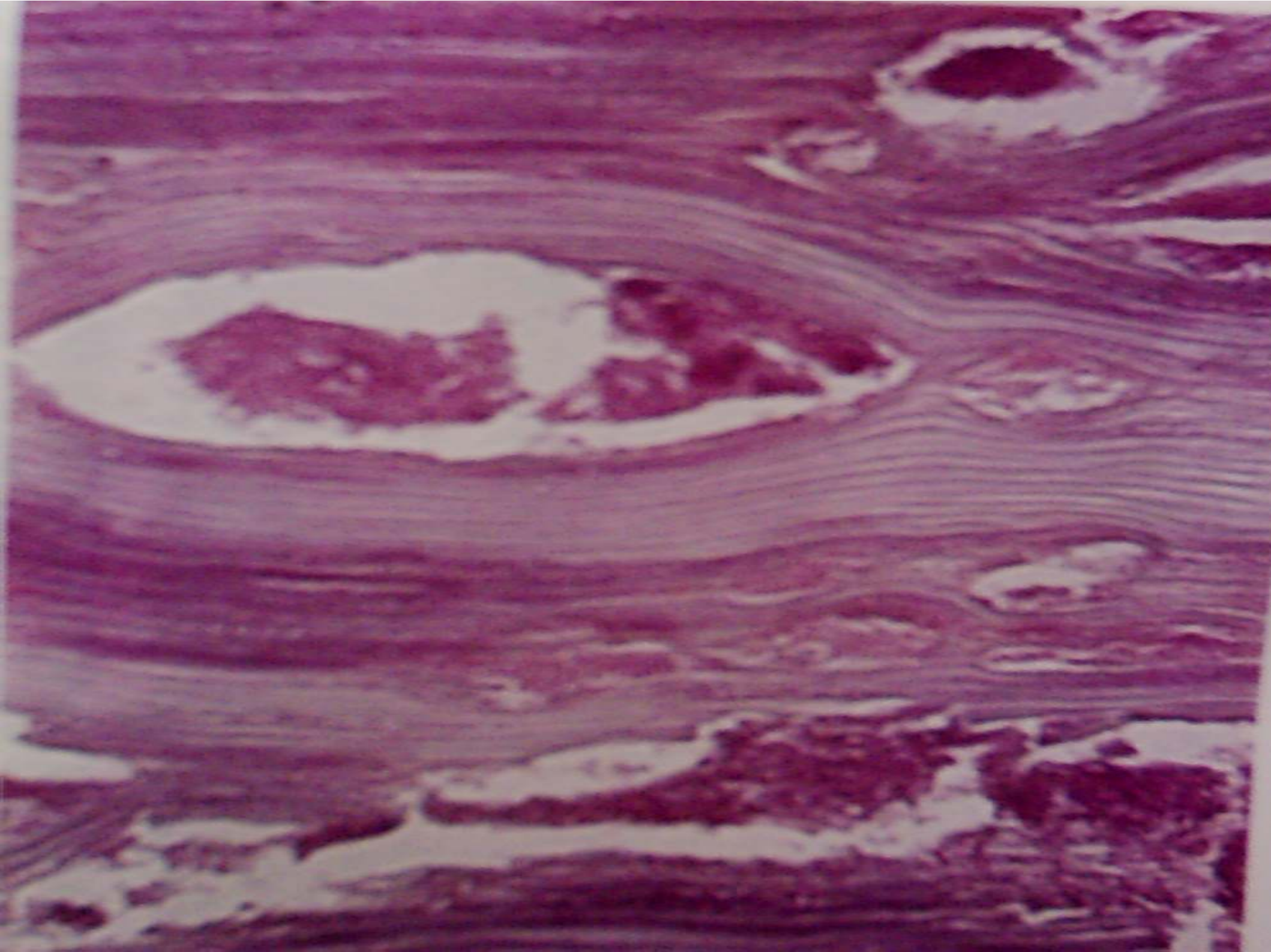
g. 2.12 Pioneer organisms at the advancing nt of the zone of bacterial invasion in dentine ries.

Fig. 2.13 Packing of the dentinal tubules by bacteria in the zone of bacterial invasion (Gram stain).

-clefts present in the carious dentin that extends at right angles to the dentinal tubules, accounts for the peeling off of dentin in layers while excavating.


Dr. Md. Nurul Islam
Dentinal cleft
Liquefaction
foci





Cemental caries

- The plaque accumulate in the cemental surface then the microorganisms penetrate the cementum along the calcified sharpies fibers then the bacteria spread vertically following the pattern of cementum formation

The cementum soften

plaque over a wide ar

producing a saucer

shaped cavity.

ROOT CARIES

- **Immunological aspects of dental caries**

Caries in man is associated with the development of serum and salivary antibodies against s. Mutans,
but in all most all individuals this natural active immunity appears to have little effect.
This may be because s. Mutans in only ***weakly antigenic***.

The salivary immune mechanism would presumably act through secretary Iga and might prevent s. Mutans from adhering to the tooth .
The Igg antibody may act as opsonin facilitating phagocytosis and ultimately death of s.mutans by neutrophils and macrophages